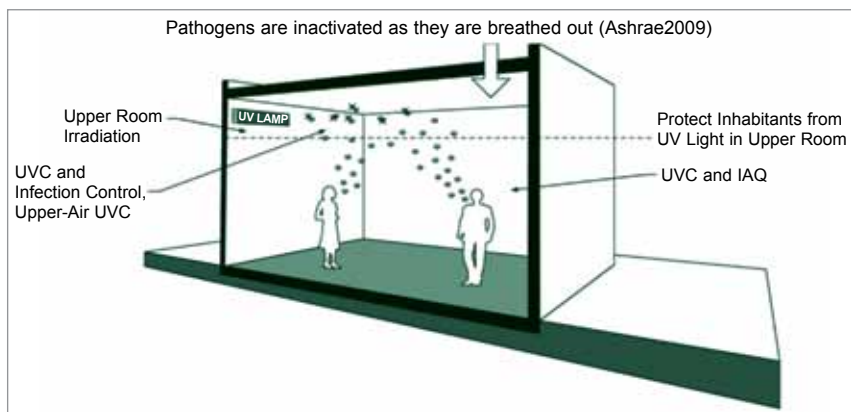




Functioning of upper-air room air UVGI (Credit: Binay Opto Electronics)

tamination, it will be safe to touch and wash with chemical disinfectants.

UV-C lamps: Competing technologies

The heart of any UVGI system is the light source, which can be from special lamps or LEDs. The most common UV light sources include mercury vapour discharge lamps, amalgam lamps, excimer lamps, and xenon lamps.

Hot-cathode, low-pressure mercury vapour lamps, which resemble fluorescent lamps without fluorescent coating, may be incorporated with UVC transmitting glass, UVC transparent soft glass, fused quartz with titanium, amalgam of mercury or indium, which blocks the mercury vapour emitting a portion of energy at 185nm, producing ozone.

Mercury lamps are currently the most cost-effective due to their high electrical to optical conversion efficiency. However, mercury lamps have a risk of toxicity. Physically, they are less robust and are also temperature sensitive, hence their monitoring is necessary. In the case of quartz glass, careful maintenance and cleaning is important. Amalgam lamps can operate at high power and have relatively longer life. Xenon lamps work in high-intensity flashes of arc discharge that deliver a large spectrum of wavelengths including UVC, UVB, UVA, and visible light. This needs filtration to remove other wavelengths; these are not commonly found in India.

A new kind of UVC lamp, the krypton chlorine excimer lamp (KRCL), has been developed that

generates UVC radiation of 222nm (aka far-UVC). It is mechanically more robust than other sources. The shorter UV wavelengths from an excimer lamp are found to be lethal to germs but do not harm the human body as they do not penetrate the upper layer. It is currently under rigorous testing and hopefully would be found safe for humans.

UVGI LEDs based on aluminium gallium nitride, which can emit UVC light in 265nm to 280nm range, are showing promising results as a viable source of UVC light emission. UVC LEDs are compact, small, and can be easily used in portable UV disinfectant solutions. They can have a high life in excess of 50,000 hours, which can be a boon for continuous use and low maintenance.

Their UVC emission efficiency is low, typically 2 to 5 percent, but unlike mercury lamps that have several lines

of emission, UVC LEDs can emit only a narrow band of emissions. However, it has the risk of exposure to the human eye as one may assume the UVC LED is turned off when it is actually on. Hence it is advised to use it only along with a secondary warning light.

The cost of UVC LEDs remains higher than mercury based light sources but is expected to come down. A recent study by researchers at University of Miyazaki (Japan) demonstrated that irradiation with a deep ultraviolet LEDs of 280nm wavelength rapidly inactivated SARS-CoV-2.

Far-UVC. Far-UVC is an emerging technology that employs different lamps to emit UV energy at a wavelength of around 222nm. It is marketed as safe for exposing humans and their direct environment to UV energy. The FDA refers to a study by scientists from the Centre for Radiological Research, Columbia University Irving Medical Centre that states far-UVC light of 222nm wavelength efficiently and safely inactivates airborne human coronavirus. The research shows far-UVC to reduce the presence of viruses, including various strains of coronavirus and influenza, up to 99.9 percent.

“Germicidal UVC light, typically at 254nm, is effective in inactivating viruses but when used directly can pose a health hazard to skin and eyes. Instead, far-UVC light (207 to 222nm) can efficiently destroy pathogens, potentially without causing harm to exposed human tissues. Continuous exposure to far-UVC in occupied public areas at the current regulatory exposure limit ($\sim 3\text{mJ}/\text{cm}^2/\text{hour}$) would cause $\sim 90\%$ viral deactivation in ~ 8 minutes, 95% in ~ 11 minutes, 99% in ~ 16 minutes, and 99.9% deactivation in ~ 25 minutes. According to current regulatory dose limits, low-dose-rate, far-UVC exposure can result in a major reduction in the ambient level of airborne coronaviruses in occupied public areas,” says Hina Gupta.

Need for standards

A key challenge inhibiting growth in adoption of UVC products is the growing confusion due to lack of standards and guidelines. In the absence of standards, it is chal-



UVGI chamber (Courtesy: www.orientelectric.com)